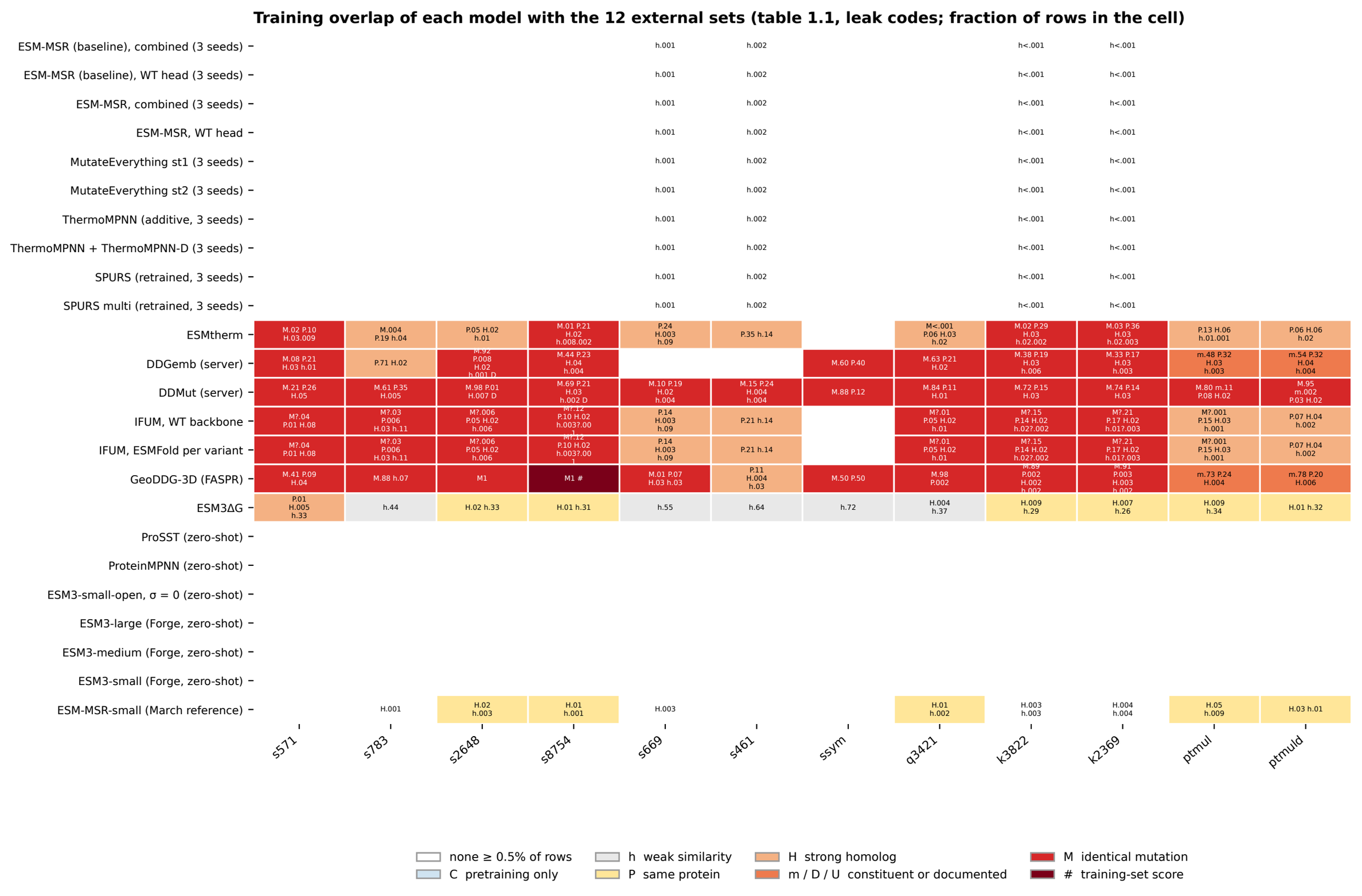
Training overlap of each model with the 12 external sets (table 1.1, leak codes; fraction of rows in the cell)
ESM-MSR (baseline), combined (3 seeds)
ESM-MSR (baseline), WT head (3 seeds)
ESM-MSR, combined (3 seeds)
ESM-MSR, WT head
MutateEverything st1 (3 seeds)
MutateEverything st2 (3 seeds)
ThermoMPNN (additive, 3 seeds)
ThermoMPNN + ThermoMPNN-D (3 seeds)
SPURS (retrained, 3 seeds)
SPURS multi (retrained, 3 seeds)
ESMtherm
DDGemb (server)
DDMut (server)
IFUM, WT backbone
IFUM, ESMFold per variant
GeoDDG-3D (FASPR)
ESM3ΔG
ProSST (zero-shot)
ProteinMPNN (zero-shot)
ESM3-small-open, σ = 0 (zero-shot)
ESM3-large (Forge, zero-shot)
ESM3-medium (Forge, zero-shot)
ESM3-small (Forge, zero-shot)
ESM-MSR-small (March reference)
s571
s783
s2648
s8754
s669
s461
ssym
q3421
k3822
k2369
ptmul
ptmuld
none ≥ 0.5% of rows
C pretraining only
h weak similarity
P same protein
H strong homolog
m / D / U constituent or documented
M identical mutation
# training-set score